ActInf Textbook Group ~ Cohort 1 ~ Meeting 8 (Chapter 4 pt. 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 8 (Chapter 4 pt. 1) | 1:00:45

Hello everyone, it's June 23rd, 2022, and we are in week 8 of the textbook group cohort 1.

We're starting chapter 4, and we'll be continuing with chapter 4 next week.

We're more than halfway done with the time and with the chapters that we're going through in the first half of the book.

So let's go to chapter 4, and just raise your hand and gather or write in the chat if you have anything that you want to address.

So I'm going first to the math overviews page, where I've written some overviews for the previous chapters of varying levels of completeness.

But this is very important as we set off into chapter 4.

So I'm page 64.

This chapter is more technical than chapters 1 through 3, appealing to linear algebra, differentiation, and a Taylor series expansion.

Those readers interested in the details may turn to the appendices, dot dot dot.

Those who do not want to delve into the theoretical underpinnings may skip this chapter.

So keep that in mind as we continue on.

And it can be an area of discussion and learning and such.

But let's approach these themes and formalisms with the authors forewarning that this is something we can look for more detail in the appendixes,

as well as skip this chapter, even if we might disagree, of course.

Okay.

Any other overall comments on chapter 4?

Just for whomever read it, even a part of it.

What was their overall perspective on chapter 4?

What was it trying to do?

What approach did they take?

What approach did they take?

What do they take?

Yeah, Ali, and then anyone else?

I think if we take the materials in chapters 1 through 3 as the foundational materials on which the active inference theory is supposed to build,

well this chapter 4 is one of the first steps toward building the actual theory and I mean going beyond just the basics and foundational materials.

So using the tools and foundations established in previous chapters, we are now perhaps ready to tackle the problem of actually constructing the generative models in two different situations as discrete time models and as continuous one.

Awesome, thanks.

Yes, in the chapters page, we can recall back to chapter 1 that just laid out the structure of the book.

Chapter 2 provided the low road to active inference, which began with Bayesian inference, and talked about a few other prerequisite or preliminary themes,

including introducing variational and expected free energy as imperatives, in the sense that they're able to be bounding surprise.

Chapter 3 introduced the high road to active inference, which was starting not from the mechanistic kind of nucleus of the Bayes equation,

but rather from the imperative for survival and persistence.

Also introducing, in a first pass, the Markov blanket concept in partitioning.

Chapter 4 is indeed when we start to get into many details that were not covered in earlier sections.

It's going to first begin by bringing us closer to this connection between Bayesian inference and the free energy evaluation.

And then this central idea of a generative model is discussed.

That is going to be described in discrete time, specifically using the POMDP formalism, partially observable Markov decision process,

and then as well as in continuous time.

Then there's going to be some very interesting figures and formalisms and discussions on what generative models underlie predictive coding and motor reflexes,

which is moving us towards Chapter 5, which is going to have some empirical work, mainly cited out,

and some discussion on the plausible neurobiologies that can be implementing or modeled as implementing or modeled with the kinds of generative models

of which have been prepared for in Chapters 2 and 3, motivated in Chapters 2 and 3 really,

and then described in their essence in Chapter 4.

Also, just a reminder that in the math group activities, but we're all in the math group, we're all in this learning journey together,

we've been striving to make the natural language descriptions for equations.

So, that would be extremely, extremely helpful.

Every annotation that people can make is helpful.

People shouldn't feel abashed or ashamed to make any kind of contribution.

It can always be re-ordered or edited by others, but this is how we're going to create those natural language descriptions of equations

and ask questions about them, even just copying and pasting.

What does this mean? What does this mean?

They're all helpful contributions.

Because this is very technical, and it's not immediately apparent how, for example,

these copious equations in Chapter 4 relate to some of the broader discussions from the earlier chapters.

But we'll get them.

Okay.

If anyone has more questions to add, they can do that.

Or if they want to upvote questions, feel free to do that.
It's motivating for other people when they see that their questions are, like, being improved
or other people are paying attention to their questions.
So, that's always something free and helpful that people can engage in.
Are beliefs policy or state?
If the person who asked it is here, they can feel free to address it.
Because it's a very short, partially formed question.
I also wasn't sure whether this was using the CAD-CAD ontology,
which uses policy and state in a slightly different way.
But beliefs, in a general sense, coming from a Bayesian ontology,
are referring to any distributional expression.
Bayesian beliefs.
Those are states.
Policies in the act in fontology,
which you can just mouse over and find out,
it's a sequence of actions.
And policies are constructed or enumerated
by the affordances, the E vector,
that the generative model has
over a given time horizon.
So, if it's able to go left or right,
over two time steps,
all the policies are left-left, left-right,
right-right, right-left.
Does anyone else just want to add anything else
to this short question?
Otherwise, it's a fine, clarifying question.
Okay.
In the discussion of active inference in POMDPs,
the authors write,
to update beliefs about policies,
we find the posterior that minimizes the free energy.
Does a posterior at time t become a prior at time t plus one?
What does anyone think?
I guess it depends on the
specific posterior and prior,
because that can mean multiple things.
But in the context of figure 4.3,

I'd say it does,
because we are updating our prior,
which becomes, at the end of a time step,
becomes the posterior probability,
which then is fed back as the prior.
Thanks.
Totally agree with that.
There needs to be an initiating prior,
d, which is shown with a 3 here,
in the discrete time.
And that is like the initial prior.
And then,
prior and posterior are referring to what?
Incoming data.
So after the data at point one comes in,
the updated posterior
is now playing the functional role of the prior
for the next data point coming in.
Okay.
Then here's sort of a related
question.
Well,
we've addressed this
question in the narrow sense,
but it would be very helpful
to look through
the order in which
different topics are addressed
in chapter 4.
Because
these are
some questions
that are invoking things
that are much later.
And
we don't assume that
people have
high or low

comprehension of this chapter
yet.
but many of these questions,
it's really important
that we understand
like why we're bringing them up.
So let's just walk through
very quickly
the chapter
to see
why they're bringing up things
in a different order.
In terms of
looking at the figures
and formalism.
4.1
is
returning to Bayes
theorem.
And
I think
this is a really nice
representation
here's the joint distribution
of X and Y.
It hidden states
in the observables
and
it's like
it gets
split up
into
Y
conditioned on X
and X.
So this is like
the probability
the joint distribution

of the coin flip
and the die
is
the distribution
of the coin flip
conditioned upon
the die
and the die itself.
So
it's just
separating out
a joint distribution
and that's
the heart
of Bayes theorem.
This is a key move
to move
an integration
problem
a sum
in the discrete case
or an integration
problem in the
continuous case
into an optimization
problem
which permits
incremental
solutions
of improving
quality
unlike an integration
problem
which
you might just
be tallying up
numbers
and not

necessarily
moving
closer
to knowing
when you're
done with
that
integral
they introduced
Jensen's
inequality
which is
the log
of the average
is always
greater or equal
to the average
of a log
and that's
true for any
shape
that has
this kind
of a curve
anything
where there's
a decreasing
curve
and here's
where that
gets applied
above
we had
this
taking this
joint distribution
multiplying it
by an arbitrary

distribution
divided by
itself
so
that's
one
whatever
q is
later q
will play
a more
specific role
and then
here
the expectation
is inside
the log
here
the expectation
is over
the log
itself
so
the one
that we'd
really want
would be
like
the joint
distribution
would be
like the
expectation
of the
joint
distribution
divided
by some

function
q
Jensen's
inequality
allows us
to
make
this
instead of
an equal
sign
make it
greater than
an equal
sign
and then
pull the
expectation
out
and utilize
that
as
the
negative
free
energy
which
is
going
to
be
bounding
us
on
this
one
that
we

would
truly
want
but
going
about
it
by
using
this
nice
feature
of the
natural
log
base
theorem
in
the
logarithmic
form
and anyone
can totally
raise their
hand
and add
details
allows
rearrangement
into
this
form
which is
already
recalling
the free
energy
that we

saw
from
variational
free
energy
from
equation
2.5
more
details
to be
filled
in
and that's
why we
want to
be
annotating
these
equations
and so
on
generative
models
is the
name
of
this
chapter
to
calculate
free
energy
we need
three
things
data
variational

distribution
family
and a
generative
model
which
at
minimum
is
composing
of the
prior
and the
likelihood
they're
going to
be
describing
two
different
kinds
of
generative
models
this is
a really
nice
and intuitive
graphic
for those
who might
have
different
familiarity
with
statistical
representations
!

The
circles
are
random
variables
and the
squares
are
probability
distributions
describing
relationships
these
are
some
Bayes
graphs
that
are
simpler
and
less
action
involved
than
this
canonical
active
representation
where pi
is policy
and so
yeah
Jakob
you wrote
three is
the B
matrix

yes
it is
there's
like
these
threes
are the
B
matrix
but
this
is
the
D
letter
so
I
agree
though
sorry
which one
is the
D
letter
the first
three
yeah
because
they're
playing
similar
roles
but
they're
a little
bit
I mean
or what

do you
think
about
that
yeah
I guess
I was
thinking
about
it
as
this
being
just
one
snapshot
of
like
an
infinite
factor
graph
so
I mean
just the
fact that
it's
assigned
the
P
of
S
tau
plus
one
given
S
tau

and
pi
I
understand
what
you
mean
by
saying
that
it's
kind
of
like
it
plays
the
role
of
the
yeah
I'm
not
entirely
sure
though
look
we'll
come back
to this
because
there's
actually
a few
other
interesting
notes
so

this
is
kind
of
the
simplest
of
the
graphs
that they
describe
this
is
like
one
variable
influencing
another
variable
this
is
like
two
factors
Z
and
X
influencing
Y
so
two
is
like
a
little
factory
that
outputs

Y
conditioned
upon
the
state
of
X
and
Z
so
knowing
this
notation
and how
to read
these
directed
graphs
is
relatively
essential
for
interpreting
more
complex
graphs
that
involve
actions
and so
on
here's
X
as like
an
upstream
hidden
cause

that
influences
two
downstream
variables
that
don't
influence
each
other
and
here
is
like
hierarchical
model
where
V
influences
X
and
X
influences
Y
does
anyone
have
thoughts
or
questions
about
this
this
is a
graphical
probabilistic
model
the

variables
are
stochastic
or
statistical
random
variables
and
it's
a
graph
consisting
of
nodes
and
edges
4.3
is going
to
introduce
these
two
basic
forms
of the
generative
model
dynamic
through
time
used
in
act
inf
in
the
factor
graph

form
does
anyone
want to
describe
what they
see
in
this
figure
the
top
is
in
the
discrete
time
case
s
is
the
hidden
state
temperature
of the
room
o
are
the
observations
the
thermometer
readings
the
two
is
the
relationship

between
the
temperature
in the
room
and the
readings
of the
thermometer
three
is
how
the
temperature
is
changing
through
time
and
then
it is
just like
Jakob
said
it is
kind
of
like
here
we
can
think
of
this
as
being
like
a

little
bit
of
a
snippet
from
this
infinite
sequence
of
temperatures
through
time
but
in
practice
there
has to
be
an
initiating
prior
pi
are
policies
policies
are
sequences
of
actions
sequences
of
affordances
that are
concatenated
over some
time
horizon

that's
what
we're
evaluating
like
expected
free
energy
on
and
policies
represent
causal
impact
in the
world
and we
can look
at this
graph
which is
why
this is
an
important
prerequisite
to understand
because the
way in which
policies
influence
the world
isn't like
by taking
the temperature
and like
just changing
it to a

different
value
it's
changing
three
which is
how the
temperature
changes
their
time
any
thoughts
on the
discrete
time
formalism
yes
mike
so as
represented in
this figure
the policy
is not
changing
over time
is that
correct
the set
of
policies
in this
figure
is not
changing
it's not
that it
can't

this is
just
only showing
two steps
of some
policy being
applied
yeah just
the way it
is in
this figure
so if
if we
were to
maybe extend
this figure
with two
more of
the bottom
section we
could also
extend
policy and
have that
changing over
time and
feeding into
that
yes and
that's actually
one of the
amazing and
interesting things
about this
graphical
representation
is we
can say

okay well
let's carry
out this
s to
five more
time steps
and then
let's have
pi one
implemented
here and
then let's
have it do
pi two
here and
it's kind
of like if
you can
draw it
you can
do it
and that's
very important
work from
devries and
fristen and
par from
around five
years ago
was demonstrating
that if
the base
graph can
be drawn
that there's
a forney
factor graph
representation

and where
there's the
forney factor
graph representation
there's
a tractable
variational
message passing
approximation
i don't know
exactly what
the guide rails
are for like
are there
graphs that
can't be
amenable to
message passing
and etc
but just
at a first
pass for
graphs that
look like
this
they can
be drawn
and more
variables can
be added
and so on
so it's
kind of like
a visual
code
or
probabilistic
graphical

models
so maybe
that's a fun
exercise is
like to think
about causal
inference in
our day
like the
temperature in
my room is
being influenced
by like whether
the air conditioner
is running in the
other room and the
temperature outside
which one of
these scenarios
does it model
onto
and just kind
of taking
scenarios that
are familiar
to us and
then separating
them in terms
of how are
the hidden
states changing
through time
what are the
observations
what policies
influence how
hidden states
change through

time
but this is
going to be
an interesting
different view
on the bottom
here with the
continuous time
model
does anyone
want to
describe the
continuous time
variance
also one
interesting
piece here is
in the live
streams that we
just did over
the last few
weeks 46
they make it
clear distinguishing
between
DAI
decision active
inference and
MAI
motor active
inference
and beyond
just modeling
cognitive
decision making
versus motor
reflex arcs
they also

highlight how
the DAI
is a discrete
time model
that's using
the POMDP
while the
motor models
tended to
use continuous
time
so they're
laying out
and also
kind of
generalizing
to show
the similarities
between
these two
different
variants
and we
can use
the notation
concordance
table
to
highlight
some of
those
parallels
this is
where the
Taylor series
approximation
comes in
and the

generalized
coordinates
of motion
though they
can also
be applied
to the
discrete
time
here
we have
x
x
prime
and x
double
prime
the
prime
notation
indicates
temporal
derivatives
and second
derivatives
if anyone
has a thought
on that
feel free
because
this is
not
doing
time
series
prediction
in the
same way

that the
POMDP
is
the
POMDP
is
in the
memory
of the
program
there's
a value
at the
previous
state
the current
state
and the
next
state
and however
many other
states
in the
time
horizon
the way
that a
continuous
time
Taylor series
approximation
is dealing
with reduction
of uncertainty
about future
data
is quite

different
than the
way that
these
discrete
time
models
are doing
so
what's
happening
here
is
the
value
of the
function
at
now
is being
estimated
or provided
then
the
first
derivative
is
calculated
so
here
3
has
the
same
structure
but
whereas
this

is
the
hidden
state
at
the
next
time
point
conditioned
upon
the
hidden
state
at
this
time
point
and
the
policy
here's
the
derivative
of
x
conditioned
upon
x
and
the
policy
v
v
also
has a
slightly
different

interpretation
because it's
not
sequence
of
events
either
and
analogously
for the
second
derivative
and the
higher
derivatives
so
they're
still
emitting
observations
but
these
are
not
observations
at
future
time
points
in the
same
way
and
that
is
visualized
in
this

box
4.2
so
here
is
x
of
t
at
time
t
so
this
is
like
the
time
series
is
going
to
go
x
of
t
is
just
this
value
here
you know
5
the
first
term
being
added
in the

Taylor
series
expansion
is
the
first
derivative
rate
of
change
at
that
x
0
that's
giving
you
a
better
approximation
through
time
then
the
second
derivative
adds
this
quadratic
feature
and
so
Taylor
series
converge
closer
and
closer

moving
further
and
further
away
from
the
target
point
as
they
include
higher
and
higher
derivatives
but
there
isn't
an
explicit
calculation
in the
Taylor
series
of
like
let's
just
say
this
is
like
1,
2,
3
time
steps

it's
not
like
what
is
going to
happen
at
3
time
steps
from
now
the
Taylor
series
could
then
be
plugged
in
with
3
to
calculate
that
however
this
is
not
calculating
x
x
prime
x
double
prime
at

t
equals
3
it's
calculating
it
for
x
sub
0
and
using
this
expansion
to
achieve
reduction
of
uncertainty
of
more
and more
distal
points
so that's
quite
interesting
and it's
again it's
an extremely
different way
of doing
time series
prediction
in the
continuous
time framework
than this

kind of
explicit
consideration
prediction
of
past
present
and
future
state
values
okay
any other
thoughts
on
4.3
because this
is one of
the most
key figures
but
the things
we can
explore
and ask
Ali
about
this
figure
on
page
69
it
says
the
relationship
between
a state

and its
temporal
derivative
here
depends
on
slowly
varying
causes
new
I just
wanted
to make
sure
does
this
mean
this
slowly
varying
causes
here
is
used
to
account
for
the
fluctuations
or
it
has
some
different
meaning
yet
slowly
varying

causes
so
policies
are
influencing
the
causal
unfolding
of
states
in this
discrete
formalization
here
in
the
continuous
time
setting
the
slowly
varying
causes
more
slowly
than
these
changes
play a
similar
role
to
policies
above
so
these
are
these

causes
intervene
in how
the
derivatives
are
calculated
no
no
I mean
here
in the
discrete
time
well
obviously
we have
we can
say
stable
policies
that
doesn't
that
don't
change
through
time
but
in
the
continuous
time
well
I don't
know
at least
that was

my
understanding
that
the
policies
can
slightly
change
but
the
change
can
be
somewhat
somehow
negligible
and
well
that's
because
of
the
additional
term
for
fluctuations
in the
equations
and
they don't
necessarily
affect
the main
components
of the
equations
one
could

set
it
up
so
that
the
policies
are
ineffectual
so
that
the
state
changes
through
time
are
with
their
own
endogenous
dynamics
not
influenced
by
policy
or
one
could
imagine
a
situation
where
the
states
don't
change
at

all
through
time
and
their
changes
are
entirely
driven
by
policy
and
I
think
analogously
there
could
be
a
setting
in
which
the
derivative
of
x
is
hardly
influenced
by
these
v
slowly
varying
causes
or
v
might

entirely
describe
the
derivatives
of
x
but
I'm not
sure if
just at
this level
of
generality
we
can
allocate
importance
to
kind of
the
endogenous
dynamics

or
policy
independent
changes
just
the
way
that
those
states
are
changing
or
analogously
derivatives

are
calculated
but
that's
a great
question
Lyle
and then
Mike
yeah
this might
be
just
slightly
off
topic
but
the
tools
if you
if you're
using a
multi-mode
simulation
tool
then
it
can
handle
both
in some
sense
it's going
to try
and solve
both
continuous
formulations

and
discrete
type
formulations
so
one
they would
typically
call
agent
based
and the
other
would be
ODE
based
or something
like that
just
the way
that
they
do
that
is
a little
bit of
sleight of
hand
because
of course
they're not
solving the
equations
per se
they're
estimating
the

solutions
and they
always
if you
have that
then if
you have
a slowly
changing
set of
parameters
like you
might
model
with the
discrete
case
then
then you
you know
you've
got a
an event
queue
and those
events
are however
far apart
they need
to be
then to
estimate
the
continuous
form
they simply
set
that

delta
t
you know
delta
t
goes
to
zero
in
some
sense
never
goes
in
the
software
exactly
to
zero
but
then
you
you
compress
that
delta
t
down
to
very
small
time
slices
and
then
you
have
estimated

your
continuous
form
and
so
by
by doing
that
you
can
really
have
a
pretty
comprehensive
way
to
represent
both
cases
in
the
same
modeling
environment
so
yes
great
point
numerical
approximations
to
continuous
processes
can
be
done
through

breaking
them
down
into
discrete
processes
and
then
making
sure
that
as
your
delta
t
is
getting
smaller
and
smaller
that
you're
getting
a
convergent
estimator
the
approach
that's
taken
analytically
here
is to
use a
Taylor
series
approximation
Mike

and
so
in
thinking
about
how
we
should
build
the
mental
model
of the
Taylor
series
approximation
or incorporate
that in
our
model
that's
one
example
of how
you might
capture
the
time
series
structure
and you
could
potentially
substitute
in
other
approaches
to

maybe
capture
finer
detail
or
discrete
events
that might
occur
on the
series
like
what
like
Loews
decomposition
or
something
like that
where you're
taking apart
components
of the
time
series
things like
trends
seasonality
discrete
events
yes
I think
the
analytical
comparisons
would be
tight
or loose

but
Kalman
filter
generalized
Bayesian
filtering
splines
time
series
decomposition
these
are all
in the
category
or like
in the
genre
of
this
Ali
by the
way
thanks
for the
clarification
that was
really
helpful
well
to ask
my question
more
explicitly
on page
78
equation
4.15
we have

some
additional
omega
terms
which
are
defined
as
stochastic
fluctuations
my question
was that
is the
term
slightly
varying
somehow
related to
these
stochastic
fluctuations
or not
yes
I believe
that the
slowly
varying
is in
relationship
to it
being included
with the
flow
component
rather than
with the
at that
timescale

stochastic
more
thermal
like
vibration
and that
was explored
in
like
live
stream
45
with
free
energy
principle
made
simpler
but not
too
simple
with this
idea
of
the
length of
n
and how
physics and
mechanics are
predicated upon
the separation
into flow
like
actions
at a
given
scale

and then
stochastic
non-flow
aligned
changes
and the
path of
least action
is when
the
limit
of
the
stochastic
term
going to
zero
but
let's
come there
as people
can see
even though
it's like
we could
totally read
this like
20 times
and it
still is
like
why is
the next
word there
why is
the next
equation
there

but
okay
they're
introducing
these
two
types
of
generative
models
that have
like
kind of
tantalizing
isomorphisms
but also
very
interesting
differences
they're
going to
first
focus
on the
discrete
time
the
partially
observable
Markov
decision
process
formalism
is used
someone
asked
like
why is

the
categorical
notation
used
it
just
allows
it
it
makes
it
easy
to look
at
a
matrix
and
to
interpret
the
matrix
as
like
a
confusion
matrix
not
just
a
matrix
that
is
confusing
like
they
can
be
but

one
that
has
to
do
with
like
a
coin
flip
so
categorical
outcomes
the
three
nodes
are the
transition
function
here
in the
categorical
context
between
different
states
that's
this
B
and
here's
where
we see
the
D
the
prior
over

the
initial
state
and
B
and
together
these
account
for the
three
nodes
in
figure
4.3
they play
a functionally
similar role
here's where
they get
distinguished
selecting
between
models
of
behavior
and
that was
another
question
which
maybe
we
could
get
to
requires
selection

amongst
these
categorically
discrete
plans
and

then
the
soft
max
normalizes
and
ensures
that
the
probability
over
those
policies
is
normalized
!
so that
they can
be
understood
as a
probability
distribution
like a
categorical
probability
distribution
here's the
expected
free energy
that we've

seen
before
more
on
expected
free
energy
specifically
on how
this
KL
divergence
term
which as
we talked
about
last week
was about
how
the
preferred
states
are
realized
active
inference
uses
f
and
g
they're
related
but they
play
different
roles
bfe
f

is the
primary
quantity
minimized
over
time
that was
interesting
to read
because
efe
is
what is
minimized
prospectively
through
time
whereas
here
the claim
is that
variational
free
energy
is
what
is
minimized
over
time
in
relationship
to
generative
model
so
again
more

more
could be
explored
there
they're
going to
focus
on a
rearrangement
of
equation
2.6
here
with
ambiguity
and
risk
being
juxtaposed
with the
informational
value
information
gain
infomax
and
pragmatic
value
and
they
then
move
into
the
linear
algebraic
form
where

the
bolds
okay
I don't
know if
bold is
used in
the same
way in
this
appendix
as it
is
in these
equations
does anyone
know about
what bold
means here
it could
mean the
vector
Ali
yes I
also think
bold
most
probably
means
vector
or
matrix
yes
it gets
pretty
subtle
though
sometimes

because
there's
times
where
there's
italics
bold
and
neutral
being
used
very
closely
it's
like
when
they're
used
closely
it can
be
confusing
and
then
when
they're
used
not
closely
it's
hard
to
juxtapose
more
expected
free
energy
rewriting

within
the
linear
algebraic
form
soft
max
bringing
back
the
logs
variational
inference
rests
upon
factorization
that's
related to
the
sparsity
of the
base
graph
Moritz
hey
so
just
going
to
410
again
I was
wondering
what's
the
use
of
the

categorical
distribution
here
because
like
maybe
I
miss
it
but
it's
not
really
explained
in
the
text
what
they
use
it
for
like
there's
one
sentence
saying
like
oh
yeah
the
fifth
line
shows
that
the
prior
belief

about
observation
is
a
categorical
distribution
like
what
is
it
for
what
does
it
do
I
I

I
never
seen
that
before
I
don't
know
if
it
has
to
be
categorical
I
like
a
preference
distribution
could

be a
continuous
function
but
here
it's
just
simpler
perhaps
to show
it as
a
categorical
like
there's
two
outcomes
having
the
food
and
not
having
the
food
so
then
that's
a
categorical
distribution
of
preferences
and
of
observations
so
they're

just
modeling
a
situation
where
there's
categorical
differences
that are
being
preferred
as
opposed
to
like
a
preference
over
some
continuous
distribution
does
that
address
it
they're
just
modeling
a
situation
where
there's
a
categorical
difference
with
observations
and

with
preferences
just
because
like
mathematically
it's
easier
now
to
start
with
that
yeah
I think
didactically
it's
a lot
clearer
because
you can
see
like
a
two
by
two
matrix
did
I
observe
the
food
or
not
did
I
get

the
food
or
not
instead
of
like
a
distribution
of
temperature
preference
and a
distribution
of
observations
I

expect
that
the
formalism
will
work
out
the
same
in
the
sense
that
you're
still
minimizing

your
surprisal
and

you're
still
performing
distribution
matching
but
also
this
is
like
a
distribution
matching
in a
categorical
context
that has
an
interpretation
of
almost
like
false
positive
and
false
negative
all
right
okay
I
will
go
with
it
!
thank
you

for

the
!
!
yeah
it's
a
good
question
and
I
mean
yeah
it
the
matrices
help
connect
to the
linear
algebra
and
the
MATLAB
representations
and
also
just
to
like
sort
of
match
or
no
match
like

in
46
the
example
had
to
do
with
like
wanting
ice
cream
and
then
they
observed
ice
cream
or
not
okay
POMDP
and also
like
where
does
this
eta
we'll
come
back
to
another
time
but
just
to
kind

of
get
through
it
all
the
first
pass
more
details
on
a
POMDP
S
and
V
auxiliary
variable
I
don't
think
this
is
the
V
here
because
we're
in
the
POMDP
setting
so
it's
just
an
auxiliary
variable

used
as
sort
of
analytical
convenience
that
concludes
their
discussion
of
the
discrete
time
model
into
continuous
time
but
first
they
describe
again
Markov
blankets
and
take
kind
of
a
second
pass
the
blankets
are
the
causes
of

X
upstream
which
are
parents
and
the
children
and
the
parents
of
the
children
the
kind
of
co
influencers
of
course
there's
more
to
say
but
that's
what
the
pearl
definition
is
how
that
gets
mapped
to
sense

and
action
states
and
what
influences
what
what's
in
the
blanket
on
the
blanket
etc
those
are
model
specific
and
then
here
are
two
common
message
passing
schemes
that
are
used
for
approximate
inference
unless
the
base
graph

is
fully
connected
there
is
a
Markov
partitioning
here
this is
pretty
funny
nb
noto bene
!
good note
good note
to note
the
slightly
non-standard
use of
the
expectation
operator
perhaps
by
overloading
it
with a
massive
subscript
but
it's
unclear
what
exactly
they meant

there
is
showing
alignment
between
these
two
different
schemes
but
there's
definitely
citations
to go
into
more
into
detail
on
belief
propagation
and
message
passing
inactive
and
then
we
see
several
of
these
figures
personally
I think
it's
slightly
challenging

to understand
whether
this is
being
used
illustratively
or
whether
these
specific
topologies
are
as
directly
interpretable
as
these
topologies
are
but
here
we see
kind of
if you
blur your
eyes
here's
policy
at the
top
here's
the
observations
here's
time
minus
one
here's

s
minus
t
minus
one
t
and
t
plus
one
so
we
can
see
some
resonances
with
4.3
but
now
we're
in
the
continuous
time
motor
active
inference
from
46
is
justifying
the
use
of
continuous
time
based

upon
the
continuous
unrolling
aspect
of
sensory
input
and
motor
output
they're
going to
start
in a
different
place
than
they
did
with
the
POMDP
model
in
this
case
we
have
much
more
of
a
physics
grounding
again
check out
live stream

number
45
to see
about
the
length
and
how
this
is
connected
but
this
is
a
much
more
physics
based
grounding
in
terms
of
like
a
flow
operator
and
a
stochastic
term
at a
given
time
scale
the
stochastic
term

is
assumed
to have
a
normal
distribution
and
that
kind
of
relates
to
what
Mike
said
about
detrending
like
one
can
think
of
this
stochastic
term
as
being
what
happens
when
you've
detrended
all
of
the
signal
that's
non

Gaussian
out
you're
left
with
like
the
residuals
which
have
a
Gaussian
nature
precision
is
capital
pi
and
that's
the
inverse
of
the
fluctuation
and
so
that's
they're
going
to
connect
that
to
the
common
filtering
here
we

return
again
to
the
generalized
coordinates
of
motion
coming
from
this
Taylor
series
approximation
route
so
if
somebody
said
I
have
10
numbers
and
we
need
to
predict
10
years
in
the
future
the
discrete
time
way
would

be
like
well
let's
try
to
predict
it
at
each
of
those
10
years
and
those
will
be
your
10
numbers
the
generalized
coordinates
of
motions
approach
which
we
explored
the
most
in
live stream
number
26
with
the

Costa
at
all
in
Bayesian
mechanics
the
10
numbers
could
be
like
the
value
today
and
the
derivative
today
the
second
derivative
and
the
third
and
fourth
and
fifth
however
many
that's
the
generalized
coordinates
of
motion
so

it's
like
a
snapshot
of
the
process
and
all
of
its
higher
derivatives
and
so
the
generalized
coordinates
of
motion
are
very
similar
to
Taylor
series
approximations
in
how
they
represent
centered
at
a
certain
point
x
naught

how
one
expects
as
movement
happens
away
from
that
x
naught
reducing
surprise
and so
here
is like
x
with a
dot
on top
is the
change
in x
and then
this is like
the derivative
of the
change
in x
and so
on
here
the
tilde
notation
is used
for the
generalized

coordinates
of
motion
so
we
see
this
equation
which
was
just
for
one
value
like
where
you are
on
the
freeway
you know
and then
this is
the
generalized
coordinates
of
motion
the
free
energy
is
written
down
for
this
generalized
coordinates

of
motion
with
precision
here
the
closest
that we
came to
exploring
it
was in
43
on
predictive
coding
there's
some more
details
on
Gaussian
the
relationship
between
Gaussian
processes
and
also
I think
Lyle
brought up
this
like
sort of
mode
and
multi-mode
simulations

this
is
modeling
a single
mode
not that
it can't
model a
bimodal
distribution
but
this is
where the
Laplace
approximation
comes into
play
this is
potentially
even
unnecessarily
complex
but
it's
there
and
the
Laplace
approximation
is
fitting
a
quadratic
distribution
that
tracks
the
mode

so
like
the
highest
point
on
the
distribution
and
then
it
fits
a
parabola
around
that
here
is
more
details
on the
hierarchical
nature
and the
kind of
multi-timescale
nature
that's
implied
ultimately
by the
Langvin
and
and
there's
a
it'd be
good to

juxtapose
figure
4-6
and
4-4
to see
how they're
similar
and different
but here
we see
message passing
happening on
the generalized
predictive
coding
architecture
and that
was like
explored a
lot more
not with
this exact
figure
but this
was
explored
more
from
like
an
analytical
and
empirical
perspective
in
43
then

there's
an
abrupt
ending
but the
following
chapters
are going
to
appeal
to the
formalisms
and apply
them
so any
thoughts
or ideas
on
chapter 4
as we
close out
and hopefully
next week
we can have
a lot of
questions and
things like
that
even basic
questions
it's like
if you read
it and you
understood it
then it'd be
awesome to
contribute a
question

that would
explore someone
else's
understanding
or prompt
towards how
you thought
about it
in a way
that made
sense for
you
and if
you don't
understand
it
then just
ask the
question
Ali and
then anyone
else
I just
wanted to
briefly mention
an additional
point related
to the
question
about
the reason
behind
using the
categorical
distribution
on page
74
it says

the fifth line
shows that
the prior belief
about the
observations
is the
categorical
distribution
whose
sufficient
statistics
are given
in the
c-vector
I think
here
sufficient
statistics
is the
key term
to understand
the reason
behind using
categorical
distributions
because
for
insufficient
statistics
we don't
necessarily
use
the exact
modeling
or distributions
instead
we use
a substitute

statistics
that is
simple enough
to calculate
and also
close enough
to the
actual
data
so
I think
that can
help
in understanding
the reason
behind that
decision
good point
and
this is
definitely
a
technical
note
but
is it
fair to
say
that
the
mean
and the
variance
are
sufficient
statistics
for a
Gaussian

distribution
yes
I think
so
so
like
that's
one of
the
perhaps
proximate
mechanisms
by which
the
Laplace
approximation
and
variational
inference
more
generally
are
able
to
deal
with
arbitrary
true
distributions
by
fitting
a
family
of
distributions
that
have
tractable

optimization
structure
and a
vastly
reduced
set
of
sufficient
statistics
if you
had
even
just
a
simple
bimodal
distribution
well
there'd
be
like
the
location
of
the
two
peaks
the
relative
heights
of
the
two
peaks
the
skewedness
and all
the

you can
have
many
many
parameters
needed
to
describe
it
in
contrast
the
Laplace
approximation
only
requires
the
location
of
the
mode
and
the
variance
estimate
and
so
where
that's
adequate
it's
simple
and fast
where
it's
inadequate
you'll
at least

know
because
you'll
continue
to be
surprised
at
new
observations
coming
in
okay
any
final
comments
on this
chapter
four
part
one
definitely
a
challenging
though
interesting
chapter
and
it
really
is
at
the
heart
of
active
and
as
we

hopefully
explore
a little
bit
today
it
includes
like
basic
intuition
pumps
as
well
as
some
Rosetta
Stone
like
representations
and
many
many
equations
which
I hope
we can
unpack
so
could
someone
explain
it
in
simple
words
it's
what
we've

been
asking
for
every
single
equation
and
it's
something
that
everyone
can
contribute
to
all
right
well
looking
forward
to
people's
edits
over
the
coming
week
and
additions
and
we'll
come
back
in
one
week
for
part
two

on

four